Lecture Notes 1

Econometrics

The Simple Linear Regression Model

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What is Econometrics?

Why Do We Need Econometrics?

Economic theory gives us *relationships*. Data lets us put *numbers* on them — and check whether they are actually true.

- By *how much* does an extra year of schooling raise a person's wage?
- Is the raw pay gap between men and women evidence of discrimination?
- If the central bank raises interest rates, how much does inflation fall — and after how long?
- Do fiscal stimulus packages actually raise output, or crowd out private spending?

Econometrics is the toolkit for answering such questions with data — and for knowing when the answers can be trusted.

What is Econometrics?

Econometrics is the unified study of economic models, mathematical statistics, and economic data.

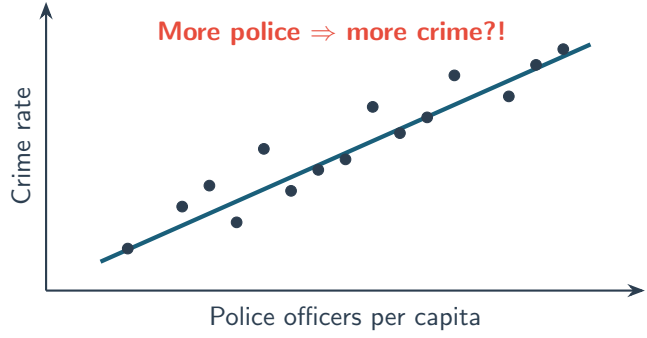
- **Theoretical econometrics** concerns the development of tools and methods, and the study of the properties of econometric methods.
- **Applied econometrics** concerns the development of quantitative economic models and the application of econometric methods to these models using economic data.

Another distinction can be made between microeconometrics and macroeconometrics.

- **Microeconometrics** is characterized by its analysis of cross section and panel data and by its focus on individual consumers, firms, and micro-level decision makers.
- **Macroeconometrics** is involved in the analysis of time-series data, usually of borad aggregates such as price levels, the money supply, exchange rates, output, investment, economic growth, and so on.

First Lesson: Correlation is Not Causation

In the data, cities with **more police** tend to have **more crime**:



Of course not — **high-crime cities hire more police**. The correlation runs backwards. Disentangling this is exactly what econometrics is for.

Data Structures in Econometrics

Cross-Sectional Data

A **cross-sectional dataset** observes N individual units at a single fixed point in time.

person_id (i)	**wage** ($/hr)	**educ** (years)	**exper** (years)	**female**	**married**
1	22.50	12	8	0	1
2	35.00	16	5	1	0
3	18.75	11	14	0	0
4	42.10	18	10	1	1
5	29.80	14	2	1	0
⋮	⋮	⋮	⋮	⋮	⋮
N	31.25	16	12	0	1

Data Structures in Econometrics

Time Series

A **time series dataset** consists of observations on a variable or several variables over time for a single entity.

time (t)	**gdp_growth** (%)	**unemp_rate** (%)	**cpi_inflation** (%)	**fed_rate** (%)
2023-Q1	2.2	3.5	5.0	4.65
2023-Q2	2.1	3.6	4.1	5.08
2023-Q3	4.9	3.7	3.7	5.33
2023-Q4	3.4	3.7	3.2	5.33
2024-Q1	1.6	3.8	3.1	5.33
⋮	⋮	⋮	⋮	⋮
T	2.8	4.1	2.6	4.83

Data Structures in Econometrics

Pooled (or Repeated) Cross Sections

A **pooled cross section** combines independent cross-sectional samples taken from different time periods.

house_id (i)	**year** (t)	**price** (\$)	**bdrms**	**sqrft**	**dist_cbd** (miles)
1	1990	145,000	3	1,600	12.5
2	1990	210,000	4	2,100	8.0
⋮	1990	⋮	⋮	⋮	⋮
120	1990	180,000	3	1,850	10.2
1	2000	225,000	3	1,500	15.0
2	2000	340,000	4	2,400	6.5
⋮	2000	⋮	⋮	⋮	⋮
135	2000	290,000	3	1,900	9.1

Data Structures in Econometrics

Panel Data

A **panel dataset** (or **longitudinal dataset**) consists of a time series for each cross-sectional member in the dataset.

county_id (i)	**year** (t)	**crime_rate**	**unemp_rate** (%)	**police_per_cap**
1	2020	45.2	6.1	2.1
1	2021	42.0	5.4	2.3
1	2022	39.8	4.8	2.5
2	2020	68.1	8.5	1.8
2	2021	65.4	7.9	1.8
2	2022	61.2	7.1	2.0
⋮	⋮	⋮	⋮	⋮
N	2022	28.4	3.9	3.1

This Pre-Course

There will be five lectures on introductory econometrics:

1. The Simple Linear Regression Model
2. The Multiple Linear Regression Model
3. Inference
4. Heteroskedasticity
5. Difference Equations

The Simple Linear Regression Model

Regression

Much of applied econometric analysis begins with the following premise: y and x are two random variables representing some population.

- We hypothesize a general relationship between y and x:

$$y = f(x) + \varepsilon$$

- $f(x)$ captures the systematic relationship between y and x.
- ε represents unobserved factors (e.g., omitted variables, measurement error) affecting y.

Simple Linear Regression

- In a simple linear regression, we simplify $f(x)$ by assuming the relationship is linear in parameters:

$$f(x) = \beta_0 + \beta_1 x \implies y = \beta_0 + \beta_1 x + \varepsilon$$

- This equation, which is assumed to hold in the population of interest, defines the simple linear regression model.
- y is called the dependent variable, explained variable, response variable, or regressand.
- x is called the independent variable, explanatory variable, control variable, predictor variable, or regressor.

Simple Linear Regression

- The variable ε is called the error term or disturbance.
- It represents factors other than x that affect y.
- A simple regression analysis treats all factors affecting y other than x as being unobserved.
- If other factors in ε are held fixed, so that a change in ε is zero, then x has a linear effect on y:

$$\Delta y = \beta_1 \Delta x \text{ if } \Delta \varepsilon = 0$$

- β_1 is the slope parameter in the relationship between y and x, holding the other factors in ε fixed.
- β_0 is called the constant term.

Example: The Keynesian Consumption Function

- Macroeconomic textbooks assume consumption depends mainly on disposable income and define the following consumption function:

$$C = c_0 + c_1 Y_D$$

- **Econometrics:** in reality, factors other than disposable income affect consumption (e.g., interest rates, wealth, consumer confidence). We express this as a regression model:

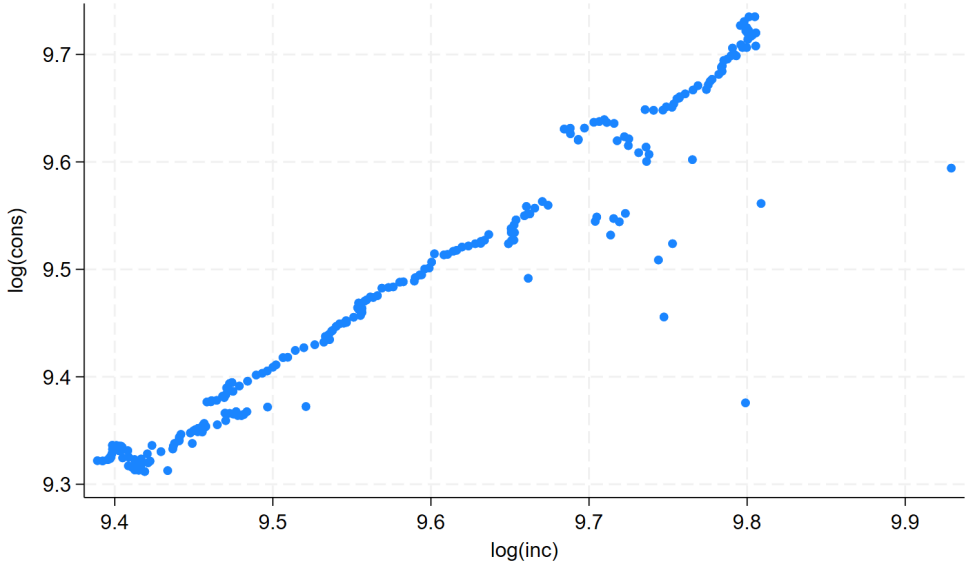
$$C = c_0 + c_1 Y_D + \varepsilon$$

where:

  - c_1 is the marginal propensity to consume,
  - c_0 is autonomous consumption.

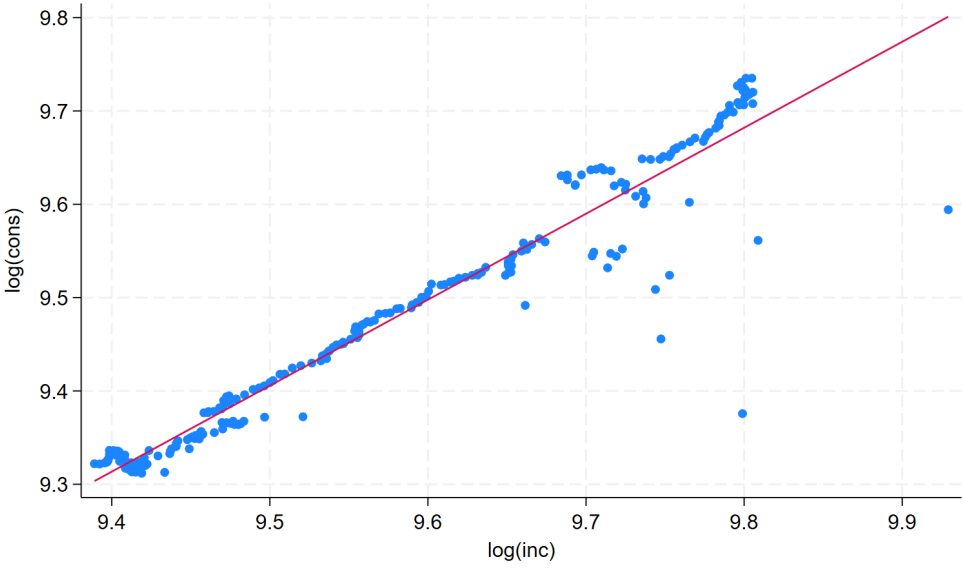
Example: The Keynesian Consumption Function

How to Estimate c_0 and c_1?



Example: The Keynesian Consumption Function

How to Estimate c_0 and c_1?



Deriving the OLS Estimates

Let $\{(x_i, y_i) : i = 1, \ldots, n\}$ denote a random sample of size n from the population.

- For each i, we can write:
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$
- To estimate β_0 and β_1, we fit a sample line $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$.
- For each i, the residual $\widehat{\varepsilon}_i$ is the vertical distance between observed y_i and predicted $\widehat{y}_i$:
$$\widehat{\varepsilon}_i = y_i - \widehat{y}_i = y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i$$
- We choose $\widehat{\beta}_0$ and $\widehat{\beta}_1$ to make the sum of squared residuals as small as possible:

$$\min_{\widehat{\beta}_0, \widehat{\beta}_1} S\left(\widehat{\beta}_0, \widehat{\beta}_1\right) = \sum_{i=1}^{n} \widehat{\varepsilon}_i^2 = \sum_{i=1}^{n} \left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i\right)^2$$

Deriving the OLS Estimates

FOCs

The first-order conditions:

$$\frac{\partial S}{\partial \widehat{\beta}_0} = -2\sum_{i=1}^{n}\left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i\right) = 0$$

$$\frac{\partial S}{\partial \widehat{\beta}_1} = -2\sum_{i=1}^{n} x_i\left(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i\right) = 0$$

giving

$$\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$$

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}$$

Sample Moments

Example: Wage and Education

A model relating a person's wage to observed education and other unobserved factors is

$$wage = \beta_0 + \beta_1 educ + \varepsilon$$

- If *wage* is measured in dollars per hour and *educ* is years of education, then β_1 measures the change in hourly wage given another year of education, holding all other factors fixed.

Using data from a sample of size $n = 526$ individuals, we obtain the following sample moments:

	Mean	Variance
wage	5.896	13.639
educ	12.563	7.667

In addition, the covariance between wages and education is 4.151.

Example: Wage and Education

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Using data from a sample of size $n = 526$ individuals, we obtain the following OLS regression line:

$$\widehat{wage} = -0.90 + 0.54 educ$$

- The slope estimate implies that one more year of education increases hourly wage by 54 cents an hour.

Algebraic Properties of OLS Statistics

1. The sum, and therefore the sample average of the OLS residuals, is zero.

$$\sum_{i=1}^{n} \widehat{\varepsilon}_i = 0$$

2. The sample covariance between the regressors and the OLS residuals is zero.

$$\sum_{i=1}^{n} x_i \widehat{\varepsilon}_i = 0$$

3. The point $(\overline{x}, \overline{y})$ is always on the OLS regression line.

Sums of Squares

For each i, write y_i as its fitted value plus its residual:

$$y_i = \widehat{y}_i + \widehat{\varepsilon}_i$$

Define the total sum of squares (TSS), the explained sum of squares (ESS), and the residual sum of squares (RSS) as:

$$TSS \equiv \sum_{i=1}^{n} (y_i - \overline{y})^2$$

$$ESS \equiv \sum_{i=1}^{n} (\widehat{y}_i - \overline{y})^2$$

$$RSS \equiv \sum_{i=1}^{n} \widehat{\varepsilon}_i^2 = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2$$

It can be shown that $TSS = ESS + RSS$.

Goodness-of-Fit

How well does the explanatory or independent variable x explains the dependent variable y?

- It is useful to compute a number that summarizes how well the OLS regression line fits the data.
- The R^2 of the regression, sometimes called the coefficient of determination, is defined as

$$R^2 \equiv \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}$$

- It is interpreted as the fraction of the sample variation in y that is explained by x.
- The value of R^2 is always between zero and one.

Example: Wage and Education

In the wage education, $R^2 = 16\%$. In addition, $ESS = 1180$, $RSS = 5981$, and $TSS = 7161$.

OLS and the Correlation Coefficient

The slope estimator is nothing more than the sample **correlation**, rescaled by the spread of the two variables:

$$\widehat{\beta}_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = \text{Corr}(x, y) \cdot \frac{s_y}{s_x}$$

where s_x and s_y are the sample standard deviations of x and y.

- The slope and the correlation always share the same **sign**: $\widehat{\beta}_1 > 0 \iff \text{Corr}(x, y) > 0$.
- Correlation is the **unit-free** version of the slope, which is why in simple regression

$$R^2 = [\text{Corr}(x, y)]^2 .$$

But correlation is not causation.

Because OLS is just rescaled correlation, it measures *association*. A nonzero $\widehat{\beta}_1$ becomes a **causal** effect only under the zero conditional mean assumption $\mathbb{E}(\varepsilon \mid x) = 0$.

The Gauss-Markov Assumptions

To establish the statistical properties of the OLS estimator, we rely on four key assumptions:

- **Zero Mean:** $\mathbb{E}(\varepsilon) = 0$
  - The expected value of the error term is zero, which means that, on average, the regression line should be correct.
- **Independence:** ε is independent of x implying $\text{Cov}(x, \varepsilon) = 0$
  - x and ε are independent.
- **Homoskedasticity:** $\text{Var}(\varepsilon) = \sigma^2$
  - All error terms have the same variance, which is referred to as homoskedasticity.
- **No serial correlation:** $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \; \forall i \neq j$
  - There is zero correlation between different error terms.

These assumptions imply that knowing x does not tell us anything about the distribution of the error terms in ε. This implies that

$$E(\varepsilon|x) = 0, \quad \text{Var}(\varepsilon|x) = \sigma^2$$

The Gauss-Markov Theorem

Under the four Gauss-Markov assumptions, the OLS estimators are **unbiased** — on average they recover the true population parameters:

$$\mathbb{E}\left(\widehat{\beta}_0\right) = \beta_0 \quad \text{and} \quad \mathbb{E}\left(\widehat{\beta}_1\right) = \beta_1$$

Proof

Moreover, among all linear unbiased estimators, they have the **smallest variance**.

The Gauss-Markov Theorem

The OLS estimators are **BLUE**: the **B**est **L**inear **U**nbiased **E**stimators.

Best smallest variance in the class of linear unbiased estimators

Linear a linear function of the sample outcomes y_i

Unbiased $\mathbb{E}(\widehat{\beta}) = \beta$

Estimators of the true parameters β_0, β_1

Variances of the OLS Estimators

- The homoskedasticity assumption plays no role in showing that $\widehat{\beta}_0$ and $\widehat{\beta}_1$ are unbiased.
- It simplifies the variance calculations.

$$\text{Var}(\widehat{\beta}_1) = \sigma^2 \frac{1}{\sum_{i=1}^{n}(x_i - \overline{x})^2} = \sigma^2 \frac{1}{TSS_x}$$

$$\text{Var}(\widehat{\beta}_0) = \sigma^2 \frac{\frac{1}{n}\sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

- However, these formulas are not usable directly, as σ^2 is unknown.
- We replace σ^2 with an unbiased estimator:

$$\widehat{\sigma^2} = \frac{1}{n-2}\sum_{i=1}^{n} \widehat{\varepsilon}^2 = \frac{RSS}{n-2}$$

- The square root of the error-variance estimator, $\widehat{\sigma} = \sqrt{\widehat{\sigma^2}}$, is called the standard error of the regression (or root mean squared error).

Variances of the OLS Estimators

Estimated Variances and Standard Errors

- Replacing σ^2 with $\widehat{\sigma^2}$ gives the **estimated variances**:

$$\widehat{\text{Var}}(\widehat{\beta}_1) = \widehat{\sigma^2} \frac{1}{\sum_{i=1}^{n}(x_i - \overline{x})^2} = \frac{\widehat{\sigma^2}}{TSS_x}$$

$$\widehat{\text{Var}}(\widehat{\beta}_0) = \widehat{\sigma^2} \frac{\frac{1}{n}\sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}$$

- Taking square roots gives the **standard errors** of the estimators:

$$\text{se}(\widehat{\beta}_1) = \frac{\widehat{\sigma}}{\sqrt{TSS_x}}, \qquad \text{se}(\widehat{\beta}_0) = \widehat{\sigma} \sqrt{\frac{\frac{1}{n}\sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n}(x_i - \overline{x})^2}}$$

- The standard errors measure the precision of the OLS estimates: smaller standard errors mean more precise estimates.
- They are the key input for hypothesis testing and confidence intervals — the subject of the next lecture on inference.

Functional Forms

Linear regression requires models to be linear in parameters, but the relationship between x and y does not need to be strictly linear in variables.

- Log-Level

$$\ln(y) = \beta_0 + \beta_1 x + \varepsilon$$

  - A 1-unit change in x changes y by $(100 \cdot \beta_1)\%$.

- Level-Log

$$y = \beta_0 + \beta_1 \ln(x) + \varepsilon$$

  - A 1% change in x changes y by $(\beta_1/100)$ units.

- Log-Log

$$\ln(y) = \beta_0 + \beta_1 \ln(x) + \varepsilon$$

  - A 1% change in x changes y by $\beta_1\%$.

Example: Log-Level Wage Equation

In economic theory, the return to education is often modeled as a percentage increase in wage rather than a fixed dollar amount:

$$\ln(wage) = \beta_0 + \beta_1 educ + \varepsilon$$

Using the sample of size $n = 526$, the estimated OLS regression line is:

$$\widehat{\ln(wage)} = 0.584 + 0.083\, educ, \quad R^2 = 0.186$$

- Each additional year of education is estimated to increase hourly wage by approximately 8.3%.

Changing the Units of Measurement

Suppose we rescale the dependent variable, e.g. measure wage in *cents* instead of dollars ($wage^* = 100 \cdot wage$).

- Both OLS estimates scale by the same factor:

$$\widehat{\beta}_0^* = 100\,\widehat{\beta}_0, \qquad \widehat{\beta}_1^* = 100\,\widehat{\beta}_1$$

- Rescaling the regressor x by a factor c instead divides the slope by c and leaves the intercept unchanged.
- In every case, the **fit is unchanged**: R^2 is invariant to the units of x and y.

This is one motivation for **logarithms**: elasticities and semi-elasticities are unit-free, so their interpretation does not depend on how the variables are scaled.

Regression on a Binary Explanatory Variable

Simple regression can also be applied to the case where x is a binary variable, often called a dummy variable, which takes on only two values, zero and one.

- Consider again the equation

$$y = \beta_0 + \beta_1 x + \varepsilon$$

- If we impose the zero conditional mean assumption, we obtain

$$\mathbb{E}(y|x) = \beta_0 + \beta_1 x + \mathbb{E}(\varepsilon|x) = \beta_0 + \beta_1 x$$

- It can be seen that

$$\mathbb{E}(y|x = 0) = \beta_0, \quad \mathbb{E}(y|x = 1) = \beta_0 + \beta_1$$

- It follows immediately that

$$\beta_1 = \mathbb{E}(y|x = 1) - \mathbb{E}(y|x = 0)$$

- β_1 is the difference in the average value of y over the subpopulations with $x = 1$ and $x = 0$.

Example: The Gender Wage Gap

Let $female = 1$ if the person is female and 0 if male. We estimate

$$wage = \beta_0 + \beta_1 \, female + \varepsilon$$

Using the same sample of $n = 526$ individuals, OLS gives:

$$\widehat{wage} = 7.10 - 2.51 \, female$$

Because *female* is binary, the estimates are just **group averages**:

- $\widehat{\beta}_0 = 7.10$ is the predicted wage when $female = 0$: the average wage for **men**, $\widehat{\mathbb{E}}(wage \mid female = 0) \approx \$7.10/\text{hr}$.
- $\widehat{\beta}_1 = -2.51$ is the **difference** in average wage between women and men:

$$\widehat{\beta}_1 = \widehat{\mathbb{E}}(wage \mid female = 1) - \widehat{\mathbb{E}}(wage \mid female = 0).$$

- Hence the average wage for **women** is $7.10 - 2.51 = \$4.59/\text{hr}$.

References

[1] Wooldridge, J. M. (2019). *Introductory Econometrics: A Modern Approach* (7th ed.). Cengage Learning.

[2] Stock, J. H., & Watson, M. W. (2019). *Introduction to Econometrics* (4th ed.). Pearson.

[3] Angrist, J. D., & Pischke, J.-S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press.

Additional Formulas

Sample Moments

Consider two variables x and y. The sample moments are given by:

1. **Sample mean**

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \quad \overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

2. **Sample variance**

$$\mathrm{Var}(x) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x}), \quad \mathrm{Var}(y) = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})$$

3. **Sample covariance**

$$\mathrm{Cov}(x, y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})$$

4. **Correlation coefficient**

$$\mathrm{Corr}(x, y) = \frac{\mathrm{Cov}(x, y)}{\sqrt{\mathrm{Var}(x)}\sqrt{\mathrm{Var}(y)}}$$

Proof: Unbiasedness of OLS

Step 1 — OLS as a linear estimator

Start from the slope estimator. Because $\sum_i (x_i - \overline{x})\,\overline{y} = 0$, the numerator simplifies and

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n}(x_i - \overline{x})^2} = \sum_{i=1}^{n} w_i\, y_i, \qquad w_i \equiv \frac{x_i - \overline{x}}{S_{xx}},$$

where $S_{xx} \equiv \sum_{i=1}^{n}(x_i - \overline{x})^2$. The weights w_i satisfy three key identities:

$$\sum_{i=1}^{n} w_i = 0, \qquad \sum_{i=1}^{n} w_i x_i = 1, \qquad \sum_{i=1}^{n} w_i^2 = \frac{1}{S_{xx}}.$$

(The first holds since $\sum(x_i - \overline{x}) = 0$; the second since $\sum(x_i - \overline{x})x_i = S_{xx}$.)

Proof: Unbiasedness of OLS

Step 2 — taking expectations

Substitute the true model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ into $\widehat{\beta}_1 = \sum_i w_i y_i$:

$$\widehat{\beta}_1 = \beta_0 \underbrace{\sum_i w_i}_{=0} + \beta_1 \underbrace{\sum_i w_i x_i}_{=1} + \sum_i w_i \varepsilon_i = \beta_1 + \sum_{i=1}^{n} w_i \varepsilon_i .$$

Under **zero mean** and **independence** (so $\mathbb{E}(\varepsilon_i \mid x) = 0$):

$$\mathbb{E}\Big(\widehat{\beta}_1\Big) = \beta_1 + \sum_i w_i \, \mathbb{E}(\varepsilon_i) = \beta_1 .$$

For the intercept, using $\widehat{\beta}_0 = \overline{y} - \widehat{\beta}_1 \overline{x}$ and $\overline{y} = \beta_0 + \beta_1 \overline{x} + \overline{\varepsilon}$:

$$\widehat{\beta}_0 = \beta_0 + (\beta_1 - \widehat{\beta}_1)\overline{x} + \overline{\varepsilon} \implies \mathbb{E}\Big(\widehat{\beta}_0\Big) = \beta_0 .$$